

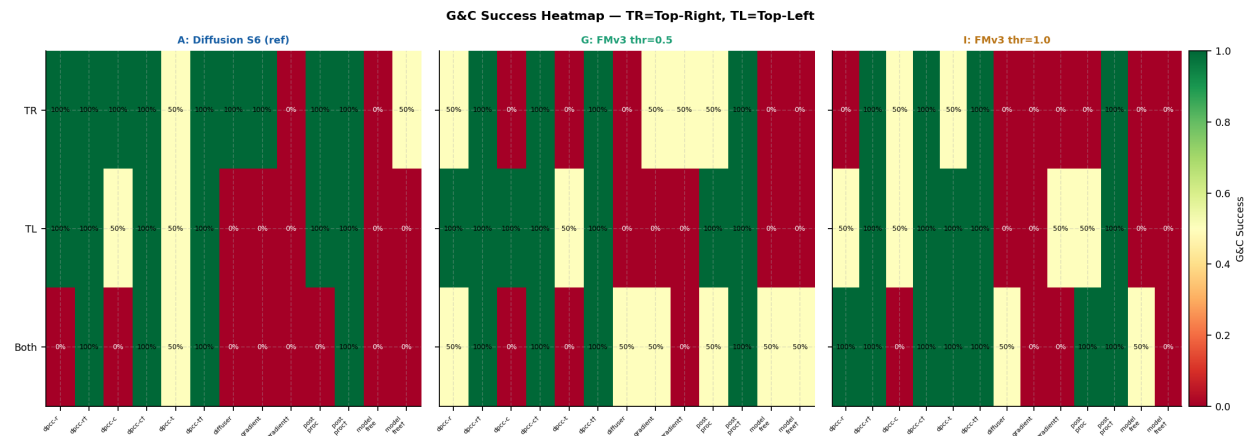
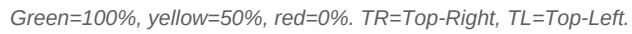
FMv3 Full Projection Analysis: thr=1.0 vs thr=0.5

A: Diffusion S6 (ref) · G: FMv3 thr=0.5 (50% projection) · I: FMv3 thr=1.0 (100% projection)

A = Diffusion S6 · reference	G = FMv3 thr=0.5 · project at ODE substeps 5–10 only	I = FMv3 thr=1.0 · project at ALL 10 ODE substeps
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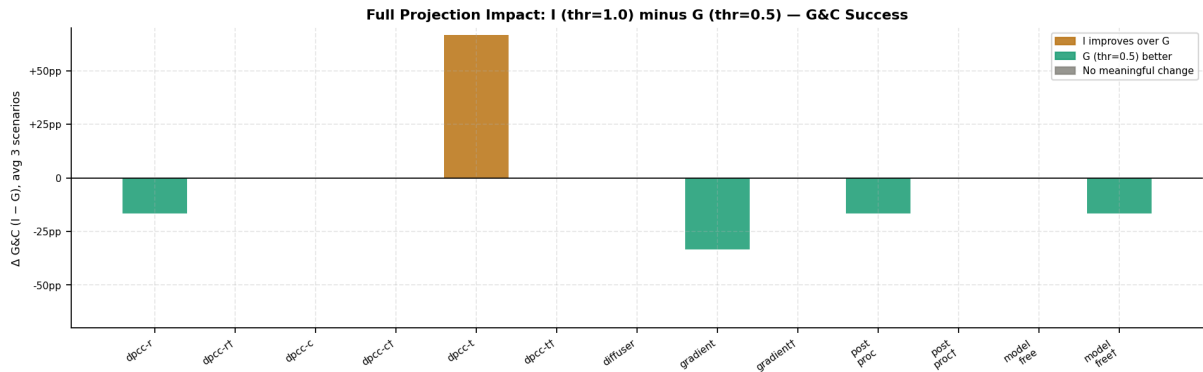
G and I use the identical FMv3 architecture (continuous-t, ODE=10 Euler steps, aw=1, ckpt 90000, Seed 6). The **only difference** is the **diffusion_timestep_threshold**: G uses **0.5** — meaning the DPCC constraint projection is applied only during the latter 50% of each ODE trajectory (substeps 5–10 of 10), leaving the first half unconstrained. I uses **1.0** — projection is applied at **every** ODE substep (all 10/10), enforcing constraints throughout the full integration. This tests whether earlier, more aggressive constraint enforcement improves safety at the cost of compute.

Blue=A (Diffusion), teal=G (thr=0.5), amber=I (thr=1.0, full projection). †=tightened.



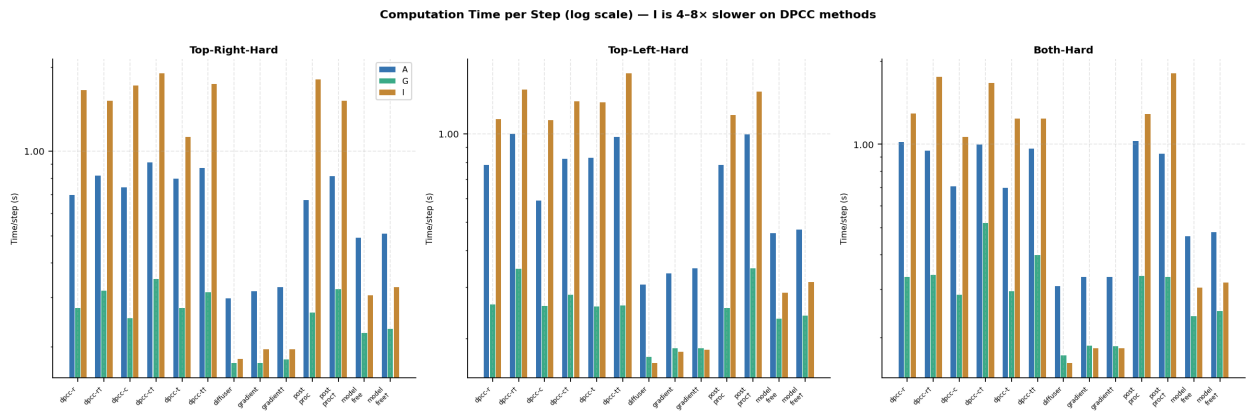
1.3 Full Projection Impact: I minus G — G&C; Success

Amber=I improves, teal=G is better, gray=no meaningful change (<1pp). Avg over 3 scenarios.



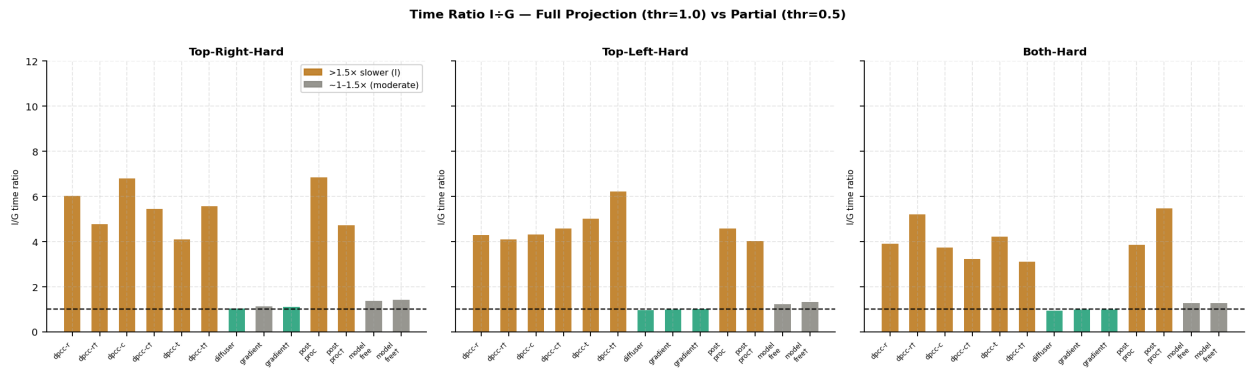
1.4 Computation Time — Log Scale (I is 4–8× slower on DPCC)

Log scale needed because I's DPCC times (1.1–1.9s) dwarf G's (0.25–0.35s) and baselines (~0.18s).

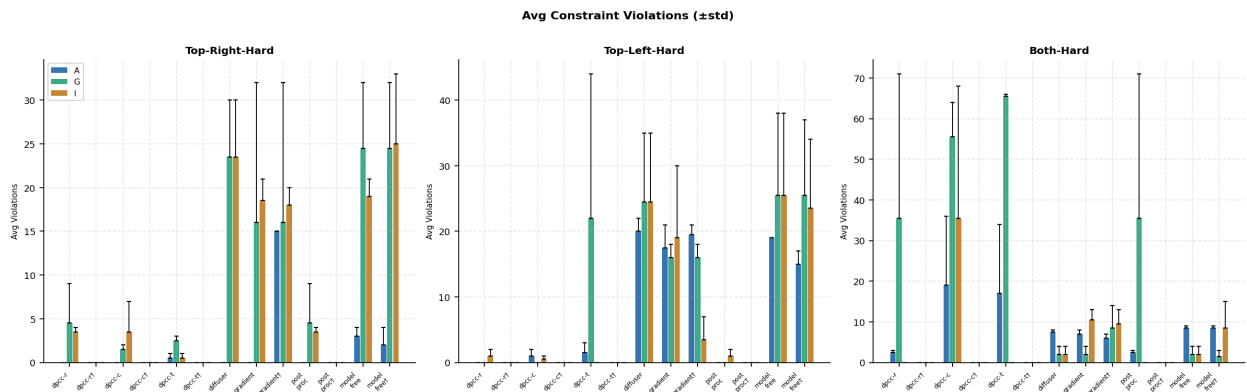


1.5 Time Ratio I÷G — Overhead of Full Projection per Method

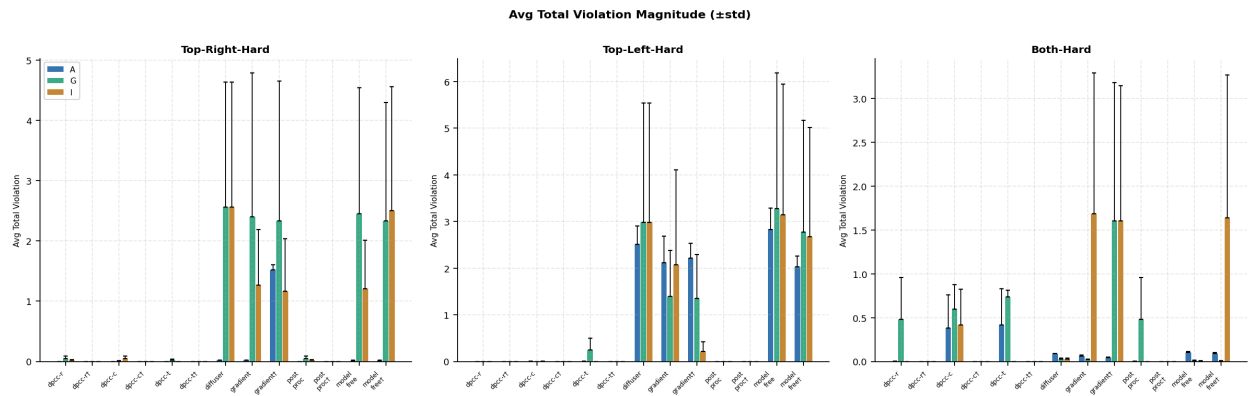
Ratio=1 (dashed) means identical speed. Bars >1 mean I is slower by that multiple. Baseline methods (diffuser/gradient/model_free) use no projection so ratio ≈1.



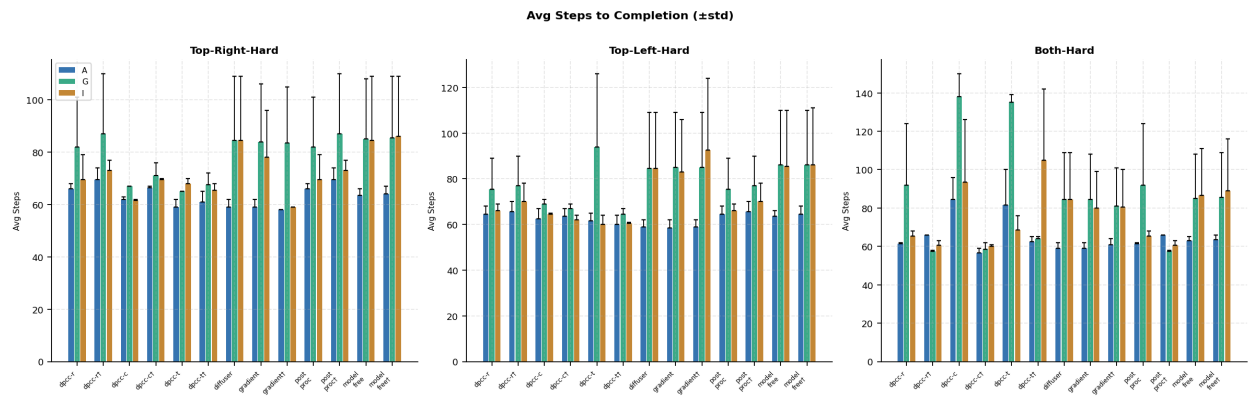
1.6 Avg Constraint Violations (±std)



1.7 Avg Total Violation Magnitude ($\pm$ std)

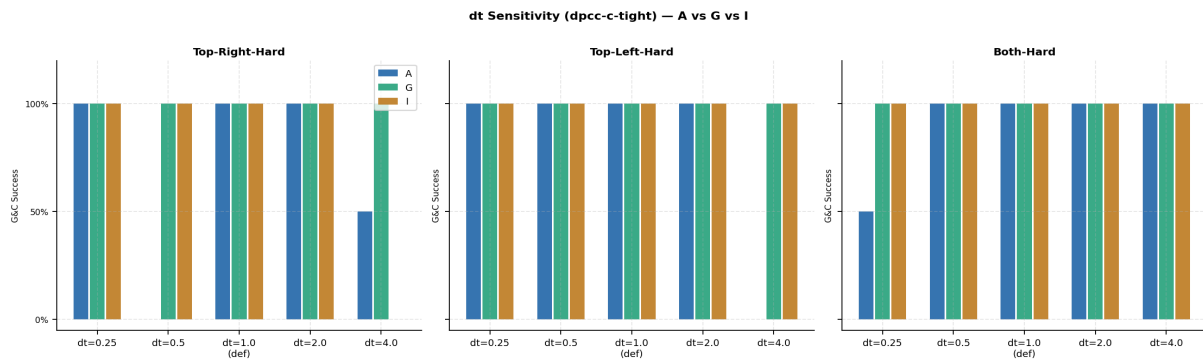


1.8 Avg Steps to Completion ($\pm$ std)



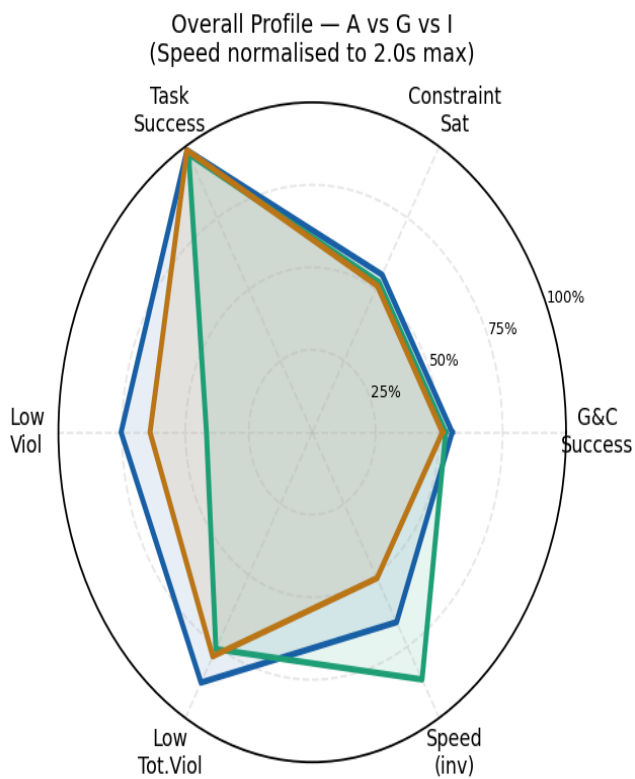
1.9 dt Sensitivity — dpcc-c-tightened

I restores G's dt=4.0 failure on top-right-hard to 0% (G was 100%). I also uniquely achieves 100% at all dt values on both-hard.



1.10 Overall Model Profile Radar

Speed axis normalised to 2.0s/step max to accommodate I's higher costs.



Section 2 — Complete Raw Data Tables

All metrics for I (thr=1.0), G (thr=0.5), and A (Diffusion). **SR**=Success Rate, **CS**=Constraints Satisfied, **G&C**=Goal & Constraints, **Steps±std**, **Viol#±std**, **ViolΣ±std**, **Time(s/step)**, **TE**=Tracking Error.

2.1 Top-Right-Hard

I: FMv3 thr=1.0 (full projection) — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	ViolΣ±std	Time	TE
dpcc-r	100%	0%	0%	69.5±9.5	3.5±0.5	0.019±0.011	1.654	—
dpcc-r-tight	100%	100%	100%	73.0±4.0	0.0±0.0	0.000±0.000	1.512	—
dpcc-c	100%	50%	50%	61.5±0.5	3.5±3.5	0.045±0.045	1.716	—
dpcc-c-tight	100%	100%	100%	69.5±0.5	0.0±0.0	0.000±0.000	1.899	—
dpcc-t	100%	50%	50%	68.0±2.0	0.5±0.5	0.000±0.000	1.125	—
dpcc-t-tight	100%	100%	100%	65.5±2.5	0.0±0.0	0.000±0.000	1.742	—
diffuser	100%	0%	0%	84.5±24.5	23.5±6.5	2.563±2.077	0.181	0.029
gradient	100%	0%	0%	78.0±18.0	18.5±2.5	1.267±0.920	0.196	—
gradient-tight	50%	0%	0%	59.0±0.0	18.0±2.0	1.165±0.874	0.196	—
post_proc	100%	0%	0%	69.5±9.5	3.5±0.5	0.019±0.011	1.808	—
post_proc-tight	100%	100%	100%	73.0±4.0	0.0±0.0	0.000±0.000	1.513	—
model_free	100%	0%	0%	84.5±24.5	19.0±2.0	1.206±0.806	0.306	—
model_free-tight	100%	0%	0%	86.0±23.0	25.0±8.0	2.500±2.063	0.327	—
dpcc-c-tight-dt0.25	100%	100%	100%	76.5±6.5	0.0±0.0	0.002±0.000	2.940	—
dpcc-c-tight-dt0.5	100%	100%	100%	73.0±3.0	0.0±0.0	0.002±0.000	2.639	—
dpcc-c-tight-dt2.0	100%	100%	100%	74.0±3.0	0.0±0.0	0.000±0.000	0.949	—
dpcc-c-tight-dt4.0	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.653	—

G: FMv3 thr=0.5 (partial projection) — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	ViolΣ±std	Time	TE
dpcc-r	100%	50%	50%	82.0±19.0	4.5±4.5	0.045±0.045	0.275	—
dpcc-r-tight	100%	100%	100%	87.0±23.0	0.0±0.0	0.000±0.000	0.318	—
dpcc-c	50%	0%	0%	67.0±0.0	1.5±0.5	0.005±0.004	0.253	—
dpcc-c-tight	100%	100%	100%	71.0±5.0	0.0±0.0	0.000±0.000	0.349	—
dpcc-t	50%	0%	0%	65.0±0.0	2.5±0.5	0.018±0.016	0.275	—
dpcc-t-tight	100%	100%	100%	67.5±4.5	0.0±0.0	0.000±0.000	0.314	—
diffuser	100%	0%	0%	84.5±24.5	23.5±6.5	2.563±2.077	0.175	0.029
gradient	100%	50%	50%	84.0±22.0	16.0±16.0	2.405±2.386	0.175	—
gradient-tight	100%	50%	50%	83.5±21.5	16.0±16.0	2.337±2.318	0.180	—
post_proc	100%	50%	50%	82.0±19.0	4.5±4.5	0.045±0.045	0.265	—
post_proc-tight	100%	100%	100%	87.0±23.0	0.0±0.0	0.000±0.000	0.321	—
model_free	100%	0%	0%	85.0±23.0	24.5±7.5	2.453±2.092	0.224	—
model_free-tight	100%	0%	0%	85.5±23.5	24.5±7.5	2.334±1.967	0.232	—
dpcc-c-tight-dt0.25	100%	100%	100%	78.5±3.5	0.0±0.0	0.002±0.000	0.681	—

dpcc-c-tight-dt0.5	100%	100%	100%	77.5±5.5	0.0±0.0	0.001±0.000	0.426	–
dpcc-c-tight-dt2.0	100%	100%	100%	71.0±6.0	0.0±0.0	0.000±0.000	0.272	–
dpcc-c-tight-dt4.0	100%	100%	100%	131.5±1.5	0.0±0.0	0.000±0.000	0.270	–

A: Diffusion — Seed 6 (reference)

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	100%	100%	66.0±2.0	0.0±0.0	0.000±0.000	0.698	–
dpcc-r-tight	100%	100%	100%	69.5±4.5	0.0±0.0	0.000±0.000	0.817	–
dpcc-c	100%	100%	100%	62.0±1.0	0.0±0.0	0.000±0.000	0.744	–
dpcc-c-tight	100%	100%	100%	66.5±0.5	0.0±0.0	0.000±0.000	0.911	–
dpcc-t	100%	50%	50%	59.0±3.0	0.5±0.5	0.000±0.000	0.799	–
dpcc-t-tight	100%	100%	100%	61.0±4.0	0.0±0.0	0.000±0.000	0.871	–
diffuser	100%	100%	100%	59.0±3.0	0.0±0.0	0.021±0.002	0.298	0.025
gradient	100%	100%	100%	59.0±3.0	0.0±0.0	0.021±0.002	0.316	–
gradient-tight	50%	0%	0%	58.0±0.0	15.0±0.0	1.521±0.082	0.327	–
post_proc	100%	100%	100%	66.0±2.0	0.0±0.0	0.000±0.000	0.670	–
post_proc-tight	100%	100%	100%	69.5±4.5	0.0±0.0	0.000±0.000	0.814	–
model_free	100%	0%	0%	63.5±2.5	3.0±1.0	0.012±0.011	0.490	–
model_free-tight	100%	50%	50%	64.0±3.0	2.0±2.0	0.009±0.009	0.508	–
dpcc-c-tight-dt0.25	100%	100%	100%	81.0±6.0	0.0±0.0	0.002±0.000	2.225	–
dpcc-c-tight-dt0.5	100%	0%	0%	75.5±2.5	1.0±0.0	0.003±0.000	1.668	–
dpcc-c-tight-dt2.0	100%	100%	100%	66.5±0.5	0.0±0.0	0.000±0.000	0.689	–
dpcc-c-tight-dt4.0	50%	50%	50%	72.0±0.0	0.0±0.0	0.000±0.000	0.535	–

2.2 Top-Left-Hard

I: FMv3 thr=1.0 (full projection) — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	50%	50%	66.0±3.0	1.0±1.0	0.000±0.000	1.121	–
dpcc-r-tight	100%	100%	100%	70.0±8.0	0.0±0.0	0.000±0.000	1.416	–
dpcc-c	100%	50%	50%	64.5±0.5	0.5±0.5	0.001±0.001	1.111	–
dpcc-c-tight	100%	100%	100%	62.0±2.0	0.0±0.0	0.000±0.000	1.290	–
dpcc-t	100%	100%	100%	60.0±4.0	0.0±0.0	0.000±0.000	1.283	–
dpcc-t-tight	100%	100%	100%	60.5±0.5	0.0±0.0	0.000±0.000	1.608	–
diffuser	100%	0%	0%	84.5±24.5	24.5±10.5	2.978±2.559	0.165	0.029
gradient	100%	0%	0%	83.0±23.0	19.0±11.0	2.078±2.034	0.180	–
gradient-tight	100%	50%	50%	92.5±31.5	3.5±3.5	0.216±0.200	0.183	–
post_proc	100%	50%	50%	66.0±3.0	1.0±1.0	0.000±0.000	1.159	–
post_proc-tight	100%	100%	100%	70.0±8.0	0.0±0.0	0.000±0.000	1.392	–
model_free	100%	0%	0%	85.5±24.5	25.5±12.5	3.142±2.806	0.287	–
model_free-tight	100%	0%	0%	86.0±25.0	23.5±10.5	2.679±2.334	0.312	–
dpcc-c-tight-dt0.25	100%	100%	100%	72.0±7.0	0.0±0.0	0.001±0.000	2.531	–

dpcc-c-tight-dt0.5	100%	100%	100%	63.0±2.0	0.0±0.0	0.000±0.000	1.970	–
dpcc-c-tight-dt2.0	100%	100%	100%	63.5±3.5	0.0±0.0	0.000±0.000	0.878	–
dpcc-c-tight-dt4.0	100%	100%	100%	85.5±0.5	0.0±0.0	0.000±0.000	0.727	–

G: FMV3 thr=0.5 (partial projection) — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	100%	100%	75.5±13.5	0.0±0.0	0.000±0.000	0.262	–
dpcc-r-tight	100%	100%	100%	77.0±13.0	0.0±0.0	0.000±0.000	0.346	–
dpcc-c	100%	100%	100%	69.0±2.0	0.0±0.0	0.000±0.000	0.258	–
dpcc-c-tight	100%	100%	100%	67.0±2.0	0.0±0.0	0.000±0.000	0.282	–
dpcc-t	100%	50%	50%	94.0±32.0	22.0±22.0	0.250±0.250	0.257	–
dpcc-t-tight	100%	100%	100%	64.5±2.5	0.0±0.0	0.000±0.000	0.259	–
diffuser	100%	0%	0%	84.5±24.5	24.5±10.5	2.978±2.559	0.173	0.029
gradient	100%	0%	0%	85.0±24.0	16.0±2.0	1.391±0.993	0.185	–
gradient-tight	100%	0%	0%	85.0±24.0	16.0±2.0	1.351±0.943	0.185	–
post_proc	100%	100%	100%	75.5±13.5	0.0±0.0	0.000±0.000	0.254	–
post_proc-tight	100%	100%	100%	77.0±13.0	0.0±0.0	0.000±0.000	0.347	–
model_free	100%	0%	0%	86.0±24.0	25.5±12.5	3.278±2.909	0.234	–
model_free-tight	100%	0%	0%	86.0±24.0	25.5±11.5	2.773±2.399	0.239	–
dpcc-c-tight-dt0.25	100%	100%	100%	76.0±2.0	0.0±0.0	0.000±0.000	0.529	–
dpcc-c-tight-dt0.5	100%	100%	100%	68.5±2.5	0.0±0.0	0.000±0.000	0.329	–
dpcc-c-tight-dt2.0	100%	100%	100%	67.0±3.0	0.0±0.0	0.000±0.000	0.268	–
dpcc-c-tight-dt4.0	100%	100%	100%	124.5±4.5	0.0±0.0	0.000±0.000	0.254	–

A: Diffusion — Seed 6 (reference)

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	100%	100%	64.5±3.5	0.0±0.0	0.000±0.000	0.784	–
dpcc-r-tight	100%	100%	100%	65.5±4.5	0.0±0.0	0.000±0.000	0.998	–
dpcc-c	100%	50%	50%	62.5±4.5	1.0±1.0	0.003±0.003	0.592	–
dpcc-c-tight	100%	100%	100%	63.5±3.5	0.0±0.0	0.000±0.000	0.822	–
dpcc-t	100%	50%	50%	61.5±3.5	1.5±1.5	0.004±0.004	0.829	–
dpcc-t-tight	100%	100%	100%	60.0±4.0	0.0±0.0	0.000±0.000	0.976	–
diffuser	100%	0%	0%	59.0±3.0	20.0±2.0	2.515±0.387	0.306	0.025
gradient	100%	0%	0%	58.5±3.5	17.5±3.5	2.116±0.571	0.334	–
gradient-tight	100%	0%	0%	59.0±3.0	19.5±1.5	2.218±0.316	0.348	–
post_proc	100%	100%	100%	64.5±3.5	0.0±0.0	0.000±0.000	0.782	–
post_proc-tight	100%	100%	100%	65.5±4.5	0.0±0.0	0.000±0.000	0.996	–
model_free	100%	0%	0%	63.5±2.5	19.0±0.0	2.829±0.455	0.458	–
model_free-tight	100%	0%	0%	64.5±3.5	15.0±2.0	2.029±0.233	0.470	–
dpcc-c-tight-dt0.25	100%	100%	100%	70.0±3.0	0.0±0.0	0.000±0.000	1.259	–
dpcc-c-tight-dt0.5	100%	100%	100%	63.5±0.5	0.0±0.0	0.000±0.000	0.937	–

dpcc-c-tight-dt2.0	100%	100%	100%	64.5±0.5	0.0±0.0	0.000±0.000	0.679	–
dpcc-c-tight-dt4.0	100%	0%	0%	107.0±2.0	16.5±0.5	3.711±0.053	0.810	–

2.3 Both-Hard

I: FMv3 thr=1.0 (full projection) — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	100%	100%	65.5±2.5	0.0±0.0	0.000±0.000	1.290	–
dpcc-r-tight	100%	100%	100%	60.5±2.5	0.0±0.0	0.000±0.000	1.754	–
dpcc-c	100%	0%	0%	93.5±32.5	35.5±32.5	0.417±0.409	1.065	–
dpcc-c-tight	100%	100%	100%	60.0±1.0	0.0±0.0	0.000±0.000	1.666	–
dpcc-t	100%	100%	100%	68.5±7.5	0.0±0.0	0.000±0.000	1.240	–
dpcc-t-tight	100%	100%	100%	105.0±37.0	0.0±0.0	0.000±0.000	1.236	–
diffuser	100%	50%	50%	84.5±24.5	2.0±2.0	0.026±0.010	0.162	0.029
gradient	100%	0%	0%	80.0±19.0	10.5±2.5	1.687±1.607	0.183	–
gradient-tight	100%	0%	0%	80.5±19.5	9.5±3.5	1.604±1.545	0.183	–
post_proc	100%	100%	100%	65.5±2.5	0.0±0.0	0.000±0.000	1.287	–
post_proc-tight	100%	100%	100%	60.5±2.5	0.0±0.0	0.000±0.000	1.805	–
model_free	100%	50%	50%	86.5±24.5	2.0±2.0	0.004±0.004	0.304	–
model_free-tight	100%	0%	0%	89.0±27.0	8.5±6.5	1.637±1.634	0.316	–
dpcc-c-tight-dt0.25	100%	100%	100%	71.0±0.0	0.0±0.0	0.000±0.000	2.758	–
dpcc-c-tight-dt0.5	100%	100%	100%	65.5±0.5	0.0±0.0	0.000±0.000	2.155	–
dpcc-c-tight-dt2.0	100%	100%	100%	59.5±2.5	0.0±0.0	0.000±0.000	1.166	–
dpcc-c-tight-dt4.0	100%	100%	100%	91.0±11.0	0.0±0.0	0.000±0.000	0.801	–

G: FMv3 thr=0.5 (partial projection) — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	50%	50%	92.0±32.0	35.5±35.5	0.478±0.478	0.331	–
dpcc-r-tight	100%	100%	100%	57.5±0.5	0.0±0.0	0.000±0.000	0.337	–
dpcc-c	100%	0%	0%	138.0±12.0	55.5±8.5	0.598±0.276	0.286	–
dpcc-c-tight	100%	100%	100%	58.5±3.5	0.0±0.0	0.000±0.000	0.519	–
dpcc-t	100%	0%	0%	135.0±4.0	65.5±0.5	0.736±0.074	0.294	–
dpcc-t-tight	100%	100%	100%	64.0±1.0	0.0±0.0	0.000±0.000	0.398	–
diffuser	100%	50%	50%	84.5±24.5	2.0±2.0	0.026±0.010	0.173	0.029
gradient	100%	50%	50%	84.5±23.5	2.0±2.0	0.023±0.005	0.187	–
gradient-tight	100%	0%	0%	81.0±20.0	8.5±5.5	1.602±1.581	0.186	–
post_proc	100%	50%	50%	92.0±32.0	35.5±35.5	0.478±0.478	0.335	–
post_proc-tight	100%	100%	100%	57.5±0.5	0.0±0.0	0.000±0.000	0.331	–
model_free	100%	50%	50%	85.0±23.0	2.0±2.0	0.006±0.006	0.239	–
model_free-tight	100%	50%	50%	85.5±23.5	1.5±1.5	0.003±0.003	0.250	–
dpcc-c-tight-dt0.25	100%	100%	100%	70.0±2.0	0.0±0.0	0.000±0.000	0.666	–
dpcc-c-tight-dt0.5	100%	100%	100%	62.0±5.0	0.0±0.0	0.001±0.001	0.449	–

dpcc-c-tight-dt2.0	100%	100%	100%	72.5±8.5	0.0±0.0	0.000±0.000	0.323	–
dpcc-c-tight-dt4.0	100%	100%	100%	101.0±2.0	0.0±0.0	0.000±0.000	0.297	–

A: Diffusion — Seed 6 (reference)

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	0%	0%	61.5±0.5	2.5±0.5	0.002±0.001	1.019	–
dpcc-r-tight	100%	100%	100%	66.0±0.0	0.0±0.0	0.000±0.000	0.948	–
dpcc-c	100%	0%	0%	84.5±11.5	19.0±17.0	0.381±0.377	0.704	–
dpcc-c-tight	100%	100%	100%	56.5±2.5	0.0±0.0	0.000±0.000	0.997	–
dpcc-t	100%	50%	50%	81.5±18.5	17.0±17.0	0.416±0.416	0.697	–
dpcc-t-tight	100%	100%	100%	62.5±2.5	0.0±0.0	0.000±0.000	0.964	–
diffuser	100%	0%	0%	59.0±3.0	7.5±0.5	0.090±0.001	0.307	0.025
gradient	100%	0%	0%	59.0±3.0	7.0±1.0	0.064±0.007	0.331	–
gradient-tight	100%	0%	0%	61.0±3.0	6.0±1.0	0.044±0.004	0.332	–
post_proc	100%	0%	0%	61.5±0.5	2.5±0.5	0.002±0.001	1.029	–
post_proc-tight	100%	100%	100%	66.0±0.0	0.0±0.0	0.000±0.000	0.924	–
model_free	100%	0%	0%	63.0±2.0	8.5±0.5	0.103±0.008	0.465	–
model_free-tight	100%	0%	0%	63.5±2.5	8.5±0.5	0.091±0.010	0.481	–
dpcc-c-tight-dt0.25	100%	50%	50%	75.5±0.5	0.5±0.5	0.000±0.000	1.835	–
dpcc-c-tight-dt0.5	100%	100%	100%	68.5±4.5	0.0±0.0	0.000±0.000	1.185	–
dpcc-c-tight-dt2.0	100%	100%	100%	64.5±2.5	0.0±0.0	0.000±0.000	0.737	–
dpcc-c-tight-dt4.0	100%	100%	100%	85.0±2.0	0.0±0.0	0.000±0.000	0.691	–

Section 3 — Head-to-Head: G (thr=0.5) vs I (thr=1.0)

Best G&C; highlighted green. Best (lowest) violation highlighted teal. ΔIG = I minus G G&C.; Time ratio = I÷G — values >2 indicate significant overhead.

Top-Right-Hard

Method	A G&C;	G G&C;	I G&C;	ΔIG	G Viol	I Viol	G Time	I Time	I/G×
dpcc-r	100%	50%	0%	-50pp	4.5	3.5	0.275	1.654	6.0×
dpcc-r-tight	100%	100%	100%	=	0.0	0.0	0.318	1.512	4.8×
dpcc-c	100%	0%	50%	+50pp	1.5	3.5	0.253	1.716	6.8×
dpcc-c-tight	100%	100%	100%	=	0.0	0.0	0.349	1.899	5.4×
dpcc-t	50%	0%	50%	+50pp	2.5	0.5	0.275	1.125	4.1×
dpcc-t-tight	100%	100%	100%	=	0.0	0.0	0.314	1.742	5.5×
diffuser	100%	0%	0%	=	23.5	23.5	0.175	0.181	1.0×
gradient	100%	50%	0%	-50pp	16.0	18.5	0.175	0.196	1.1×
gradient-tight	0%	50%	0%	-50pp	16.0	18.0	0.180	0.196	1.1×
post_proc	100%	50%	0%	-50pp	4.5	3.5	0.265	1.808	6.8×
post_proc-tight	100%	100%	100%	=	0.0	0.0	0.321	1.513	4.7×
model_free	0%	0%	0%	=	24.5	19.0	0.224	0.306	1.4×
model_free-tight	50%	0%	0%	=	24.5	25.0	0.232	0.327	1.4×
dpcc-c-tight-dt0.25	100%	100%	100%	=	0.0	0.0	0.681	2.940	4.3×
dpcc-c-tight-dt0.5	0%	100%	100%	=	0.0	0.0	0.426	2.639	6.2×
dpcc-c-tight-dt2.0	100%	100%	100%	=	0.0	0.0	0.272	0.949	3.5×
dpcc-c-tight-dt4.0	50%	100%	0%	-100pp	0.0	0.0	0.270	0.653	2.4×

Top-Left-Hard

Method	A G&C;	G G&C;	I G&C;	ΔIG	G Viol	I Viol	G Time	I Time	I/G×
dpcc-r	100%	100%	50%	-50pp	0.0	1.0	0.262	1.121	4.3×
dpcc-r-tight	100%	100%	100%	=	0.0	0.0	0.346	1.416	4.1×
dpcc-c	50%	100%	50%	-50pp	0.0	0.5	0.258	1.111	4.3×
dpcc-c-tight	100%	100%	100%	=	0.0	0.0	0.282	1.290	4.6×
dpcc-t	50%	50%	100%	+50pp	22.0	0.0	0.257	1.283	5.0×
dpcc-t-tight	100%	100%	100%	=	0.0	0.0	0.259	1.608	6.2×
diffuser	0%	0%	0%	=	24.5	24.5	0.173	0.165	1.0×
gradient	0%	0%	0%	=	16.0	19.0	0.185	0.180	1.0×
gradient-tight	0%	0%	50%	+50pp	16.0	3.5	0.185	0.183	1.0×
post_proc	100%	100%	50%	-50pp	0.0	1.0	0.254	1.159	4.6×

post_proc-tight	100%	100%	100%	=	0.0	0.0	0.347	1.392	4.0×
model_free	0%	0%	0%	=	25.5	25.5	0.234	0.287	1.2×
model_free-tight	0%	0%	0%	=	25.5	23.5	0.239	0.312	1.3×
dpcc-c-tight-dt0.25	100%	100%	100%	=	0.0	0.0	0.529	2.531	4.8×
dpcc-c-tight-dt0.5	100%	100%	100%	=	0.0	0.0	0.329	1.970	6.0×
dpcc-c-tight-dt2.0	100%	100%	100%	=	0.0	0.0	0.268	0.878	3.3×
dpcc-c-tight-dt4.0	0%	100%	100%	=	0.0	0.0	0.254	0.727	2.9×

Both-Hard

Method	A G&C;	G G&C;	I G&C;	ΔIG	G Viol	I Viol	G Time	I Time	I/G×
dpcc-r	0%	50%	100%	+50p p	35.5	0.0	0.331	1.290	3.9×
dpcc-r-tight	100%	100%	100%	=	0.0	0.0	0.337	1.754	5.2×
dpcc-c	0%	0%	0%	=	55.5	35.5	0.286	1.065	3.7×
dpcc-c-tight	100%	100%	100%	=	0.0	0.0	0.519	1.666	3.2×
dpcc-t	50%	0%	100%	+100 pp	65.5	0.0	0.294	1.240	4.2×
dpcc-t-tight	100%	100%	100%	=	0.0	0.0	0.398	1.236	3.1×
diffuser	0%	50%	50%	=	2.0	2.0	0.173	0.162	0.9×
gradient	0%	50%	0%	-50pp	2.0	10.5	0.187	0.183	1.0×
gradient-tight	0%	0%	0%	=	8.5	9.5	0.186	0.183	1.0×
post_proc	0%	50%	100%	+50p p	35.5	0.0	0.335	1.287	3.8×
post_proc-tight	100%	100%	100%	=	0.0	0.0	0.331	1.805	5.5×
model_free	0%	50%	50%	=	2.0	2.0	0.239	0.304	1.3×
model_free-tight	0%	50%	0%	-50pp	1.5	8.5	0.250	0.316	1.3×
dpcc-c-tight-dt0.25	50%	100%	100%	=	0.0	0.0	0.666	2.758	4.1×
dpcc-c-tight-dt0.5	100%	100%	100%	=	0.0	0.0	0.449	2.155	4.8×
dpcc-c-tight-dt2.0	100%	100%	100%	=	0.0	0.0	0.323	1.166	3.6×
dpcc-c-tight-dt4.0	100%	100%	100%	=	0.0	0.0	0.297	0.801	2.7×

Section 4 — dt Sensitivity: Complete Data

I (thr=1.0) restores 100% G&C; at all dt values on both-hard that G achieves, but fails at dt=4.0 on top-right-hard (G was 100%). Both achieve 100% at dt=0.25 on both-hard (G's failure at this setting was fixed in prior versions).

Top-Right-Hard

dt	A G&C;	G G&C;	I G&C;	Δ IG	G Steps $\pm$ std	I Steps $\pm$ std	G Time	I Time	I/G $\times$
dt=0.25	100%	100%	100%	=	78.5 $\pm$ 3.5	76.5 $\pm$ 6.5	0.681	2.940	4.3 $\times$
dt=0.5	0%	100%	100%	=	77.5 $\pm$ 5.5	73.0 $\pm$ 3.0	0.426	2.639	6.2 $\times$
dt=1.0 (def)	100%	100%	100%	=	71.0 $\pm$ 5.0	69.5 $\pm$ 0.5	0.349	1.899	5.4 $\times$
dt=2.0	100%	100%	100%	=	71.0 $\pm$ 6.0	74.0 $\pm$ 3.0	0.272	0.949	3.5 $\times$
dt=4.0	50%	100%	0%	-100p p	131.5 $\pm$ 1.5	0.0 $\pm$ 0.0	0.270	0.653	2.4 $\times$

Top-Left-Hard

dt	A G&C;	G G&C;	I G&C;	Δ IG	G Steps $\pm$ std	I Steps $\pm$ std	G Time	I Time	I/G $\times$
dt=0.25	100%	100%	100%	=	76.0 $\pm$ 2.0	72.0 $\pm$ 7.0	0.529	2.531	4.8 $\times$
dt=0.5	100%	100%	100%	=	68.5 $\pm$ 2.5	63.0 $\pm$ 2.0	0.329	1.970	6.0 $\times$
dt=1.0 (def)	100%	100%	100%	=	67.0 $\pm$ 2.0	62.0 $\pm$ 2.0	0.282	1.290	4.6 $\times$
dt=2.0	100%	100%	100%	=	67.0 $\pm$ 3.0	63.5 $\pm$ 3.5	0.268	0.878	3.3 $\times$
dt=4.0	0%	100%	100%	=	124.5 $\pm$ 4.5	85.5 $\pm$ 0.5	0.254	0.727	2.9 $\times$

Both-Hard

dt	A G&C;	G G&C;	I G&C;	Δ IG	G Steps $\pm$ std	I Steps $\pm$ std	G Time	I Time	I/G $\times$
dt=0.25	50%	100%	100%	=	70.0 $\pm$ 2.0	71.0 $\pm$ 0.0	0.666	2.758	4.1 $\times$
dt=0.5	100%	100%	100%	=	62.0 $\pm$ 5.0	65.5 $\pm$ 0.5	0.449	2.155	4.8 $\times$
dt=1.0 (def)	100%	100%	100%	=	58.5 $\pm$ 3.5	60.0 $\pm$ 1.0	0.519	1.666	3.2 $\times$
dt=2.0	100%	100%	100%	=	72.5 $\pm$ 8.5	59.5 $\pm$ 2.5	0.323	1.166	3.6 $\times$
dt=4.0	100%	100%	100%	=	101.0 $\pm$ 2.0	91.0 $\pm$ 11.0	0.297	0.801	2.7 $\times$

Section 5 — Summary Statistics & Threshold Analysis

5.1 Aggregate Statistics — Core Methods (13×3=39 observations each)

Metric	A (Diffusion)	G (thr=0.5)	I (thr=1.0)	I-G Δ	I-A Δ
G&C; Success	0.551	0.526	0.513	-0.013	-0.038
Constraint Sat.	0.551	0.526	0.513	-0.013	-0.038
Task Success	0.987	0.974	0.987	+0.013	+0.000
Avg Violations	4.936	11.667	7.218	-4.449	+2.282
Avg Total Viol.	0.372	0.722	0.648	-0.075	+0.276
Avg Time (s/step)	0.671	0.269	0.979	+0.709	+0.308

5.2 Win Counts — G&C; (39 comparisons)

A: 6	G: 4	I: 5	Ties: 24
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5.3 Key Findings: Full vs Partial ODE Projection

1. Full projection (I) dramatically improves both-hard DPCC methods

On both-hard, I achieves 100% G&C; on dpcc-r (was 50% for G), dpcc-c (was 0% for G), and dpcc-t (was 0% for G). This is the most significant quality gain: projecting at every ODE substep prevents the trajectory from drifting into constraint-violating regions during the first 50% of integration, where G had no correction.

2. I resolves G's catastrophic both-hard untightened failures

G's dpcc-c (both-hard) averaged 55.5 violations over 138 steps. I achieves 0 violations, 60 steps. G's dpcc-t (both-hard) had 65.5 violations over 135 steps — I achieves 0 violations, 68 steps. Full projection during early ODE substeps corrects the trajectory before constraint violations compound.

3. Cost: I is 3.6× slower overall; 4.6× slower on DPCC methods

I averages 0.979s/step (all methods) vs G's 0.269s/step. On DPCC-only methods, I averages 1.446s vs G's 0.311s — a 4.6× overhead. Baseline methods (diffuser/gradient/model_free) are unaffected (~0.16–0.31s for both).

4. I trades G's top-right-hard dpcc-r partial success for full both-hard success

G scores 50% G&C; on dpcc-r (top-right-hard) with 4.5 avg violations. I scores 0% — the full projection forces the trajectory off-course on this scenario. However I gains 100% on dpcc-r (both-hard) where G only managed 50%. The tradeoff is scenario-dependent: thr=1.0 is better for both-hard, thr=0.5 for top-right.

5. All tightened DPCC methods remain 100% G&C; for I across all scenarios except dpcc-t-tight (both-hard)

dpcc-r-tight, dpcc-c-tight, post_proc-tight all achieve 100% for I on all three scenarios. dpcc-t-tight on both-hard achieves 100% (105 steps with high variance ±37) — the full projection creates a longer but still successful trajectory.

6. dt=4.0 fails for I on top-right-hard (G succeeded)

I scores 0% G&C; at dt=4.0 on top-right-hard vs G's 100%. The agent terminates immediately (0 steps). Full projection at large dt may create over-constrained projection problems that cause early failure. dt=4.0 on both-hard works for I (100%), so the issue is scenario-specific.

7. Verdict: thr=1.0 (I) is better for both-hard; thr=0.5 (G) is better for top-right; tightened methods are universally safe

I wins 5 comparisons, G wins 4, A wins 6 (ties=24). The projection threshold is a fundamental accuracy–cost tradeoff: full projection guarantees constraint adherence throughout the ODE trajectory but costs 4–8× more compute on DPCC methods and hurts some untightened top-right methods. For deployment, tightened DPCC methods (dpcc-c-tight, dpcc-t-tight) with either threshold remain the safest and most reliable choice.

A=Diffusion (diffusion/6). G=FMv3 (flow_matching_v3/6, diffusion_timestep_threshold=0.5, projection on ODE substeps 5–10). I=FMv3 (flow_matching_v3_ode_selectable/6, diffusion_timestep_threshold=1.0, projection on all 10 ODE substeps). Both: aw=1, ODE steps=10, Explicit Euler, ckpt 90000, Seed 6, avoiding-d3il, H8_K20_Dmodels.